

Ex. 1: Write each of these predicates in Prolog by filling in the blanks.

a) [5 pts] **last(L, X)** is satisfied when X is the final element of list L.

Example: last([2,3,4,5,6,7], X) yields X = 7.

last([X], X).

last([_ | T], X) :- last(T, X).

b) [5 pts] **removelast(L, L2)** is satisfied when L2 is formed from list L by discarding the final element of L. Example: removelast([2,3,4,5,6,7], Y) yields Y = [2,3,4,5,6].

removelast([_], []).

removelast([H | T], [H | T2]) :- removelast(T, T2).

Ex.2: Consider the following predicates:

<pre>plays(flagTwirlers, flags). plays(plankton, keyboard). plays(squidward, clarinet). plays(patrick, mayo). plays(patrick, drums). plays(spongebob, drums). plays(spongebob, spatula).</pre>	<pre>twoPlayers(X) :- plays(Y, X), plays(Z, X), Y \= Z. talented(X) :- plays(X, Y), plays(X, Z), Y \= Z. band1([], []). band1([H T], [X Y]) :- talented(H), plays(H, X), band1(T, Y).</pre>
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For each query, state whether the query is true or list all of the substitution(s) that make the query true. If the query is false or if there are no substitutions, write false .

a) ?- plays(patrick,X).

X = mayo;

X = drums.

b) ?- twoPlayers(X).

X = drums;

X = drums.

c) ?- talented(squidward).

False.

d) ?- band1([patrick, flagTwirlers], X).

False.

e) ?- band1(X, [drums, mayo]).

X = [patrick, patrick]